

Chapter 1 — Section 1.6: Data Science and AI Components

Digital Twin Cyber Deception System (LabyrinthAI)

Drafted by Abdelrahman Mohamed (692300146) — Data & Feature Engineering & LLM Lead Covers 1.6.1, 1.6.2, 1.6.3. Sections 1.6.4-1.6.6 are Seif Mohamed.

1.6 Data Science and Artificial Intelligence Components

LabyrinthAI uses data for two opposite purposes:

- **Real data teaches the system.** Recorded network traffic, already marked as normal or attack, trains the classifier to tell legitimate users from attackers, and novices from skilled attackers.
- **Fake data deceives the attacker.** Generated emails, documents, and records fill the Digital Twin so it looks like the real network.

A third source appears once the Digital Twin — an isolated replica of the corporate network, built from fake servers and files, into which attackers are redirected instead of reaching the real ones — is running. Every attacker who enters is recorded: the commands they run, the files they open, the time between their actions. One attacker's visit produces one **session log**. These logs are the project's own data, collected from its own system, and are used to improve the classifier beyond what a public dataset from 2018 can teach it.

1.6.1 DATASET DESCRIPTION AND SOURCE

Dataset	Source / Owner	Size / Records	Features / Data Type	Target / Labels	License / Access
CSE-CIC-IDS 2018	CSE and the Canadian Institute for Cybersecurity, University of New Brunswick	≈16M connection records, 10 days; ≈6.5 GB CSV	80 measurements per connection, extracted by CICFlowMeter-V3; raw packet captures; host logs	Label : Benign, or one of 7 attack types	Free for academic use; citation required
Digital Twin session logs	Collected by the project through Wazuh/ELK	Grows from Phase 2 onward	Commands run, files opened, timing, movement between machines	Attacker skill level, set by the project's own attack team	Project-owned; isolated lab
Decoy corpus	Generated by the project using Llama 3 (via Ollama) and Faker	Generated on demand	Emails, reports, employee and finance tables	None — this is output, not training input	Project-owned; fully invented

Why this dataset. KDD Cup 99 and NSL-KDD describe networks that no longer exist. DARPA 98/99 was generated artificially rather than captured. CSE-CIC-IDS 2018 was recorded from a real victim organisation of five departments, 420 client machines, and 30 servers, attacked by 50 machines running tools still in use. Its attack types — brute force, DoS, DDoS, web attack, infiltration, botnet — are the same behaviours LabyrinthAI must recognise.

Known limitations. The dataset was checked, not assumed. Four flaws shape the cleaning steps in Section 1.6.3:

- **Class imbalance.** Most records are normal traffic. Some attacks appear only tens of times. A model can therefore score highly while catching nothing.
- **Label quality.** Engelen et al. (2021) found records marked with the wrong attack type, mostly in the Infiltration class. Each attack type is checked against the schedule the authors published; any type that cannot be trusted is excluded, and the exclusion stated.
- **File defects.** Column headings repeat in the middle of some files. Some records are exact copies of others. Two speed columns, `Flow Bytes/s` and `Flow Packets/s`, contain `Infinity` and `NaN` where a connection lasted no measurable time.
- **Environment specificity.** Addresses, port numbers, and timestamps describe the one network that was recorded, not how attackers behave. Section 1.6.3 removes them.

1.6.2 DATA COLLECTION METHOD

Collection Aspect	Planned Method / Decision
Source and acquisition	• Public dataset — download the ready-made CSV files from CIC. Keeps the measurements identical to the studies compared in Chapter 2. • Session logs — collected by Wazuh/ELK from every fake service in the Digital Twin. • Fake content — generated on project machines. No outside service is called.
Sampling / coverage	• All ten recording days are kept, so every attack type is present. • Normal traffic is reduced only in the training portion, by the step in Section 1.6.3.
Labeling / annotation	• Public dataset — labels come from the schedule the authors followed when running their attacks. Checked before use. • Session logs — labelled by the Red Team, the project members who play the attacker. They run each attack, so they know what happened. • Fake content — needs no labels.
Permissions / ethics	• Public dataset — published for academic use, cited. Recorded from a laboratory organisation, so no personal data. • Our own recordings — isolated lab, machines the team owns, attacked by the team. No live network, outside company, or real person. • Fake content — every name and figure is invented.
Known collection bias	• Age — tools are from 2018, so newer techniques are missing. Our own session logs cover them. • Environment — one network was recorded. Features describing it are removed. • Operator — a small Red Team attacks in a recognisable style. Attacks are scripted from standard tools, and the operator is rotated.

1.6.3 DATA PREPROCESSING

Everything the pipeline learns — scaling averages, encodings, resampling — is calculated on the training portion alone, then applied unchanged to validation and test.

Stage	Planned Operation	Reason / Leakage Control
Data quality checks	• Remove repeated column headings. • Delete records that are exact copies. • Replace <code>Infinity</code> and <code>NaN</code> in the two speed columns. • Check record counts against the published schedule.	• Duplicates go before the split, or the same record lands in both training and test. • The non-finite values come from connections of zero length and corrupt every later calculation.
Cleaning / transformation	• Remove columns that identify machines rather than behaviour: source and destination IP address, source and destination port, <code>Flow ID</code>, <code>Timestamp</code>. • Remove columns holding the same value in every record. • Make attack names consistent.	• Keep the attacker's IP and the model memorises the machine, not the attack. It scores near-perfectly in testing and fails against any attacker it has not seen. • Keep <code>Timestamp</code> and it learns the attack schedule instead of the attacks.
Feature engineering	• Keep measurements that describe behaviour: connection length, packets and bytes each way, gaps between packets, packet sizes, TCP flag counts. • Add two ratios — packets sent versus received, and average bytes per packet. • Where two measurements always rise and fall together, keep the one easier to explain.	• Behaviour transfers to the Digital Twin. Raw volumes describe the recorded network. • SHAP/LIME must name the measurement behind a decision, so measurements stay explainable.
Scaling / encoding	• Rescale numerical measurements, using averages from the training portion only. • Save those values and reuse them live, so live traffic is treated like training traffic. • Convert attack names to	• Averages taken from all the data leak the test set's shape into training.

Stage	Planned Operation	Reason / Leakage Control
	numbers. • Use the tokenizer that ships with Llama 3 for generated text.	
Imbalance / augmentation	• Reduce the number of normal records. • Create extra examples of rare attacks using SMOTE. • Give rare attacks more weight during training. • Merge or drop attack types with too few records, and say which were dropped.	• Resampling before the split puts near-copies of training rows into the test set. • Validation and test keep the real mix, or the false-alarm rate in Section 1.6.6 means nothing.

Decoy corpus quality gate. Generated content is what the system produces, not what it learns from, so it is checked differently:

- **Internal consistency.** An employee named in a fake email must exist in the fake employee list. File paths must lead somewhere. One contradiction tells the attacker the environment is a trap.
- **No real data leaked.** Content is checked for personal details the language model may have reproduced from its own training data.
- **No credential collision.** No invented password may match a real one used anywhere in the project.

REFERENCES FOR THIS SECTION

Merge into Section 1.5.4, IEEE style:

- I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization," in *Proc. IC/ISSP*, Funchal, Portugal, 2018, pp. 108–116.
- Canadian Institute for Cybersecurity, "CSE-CIC-IDS2018 on AWS," University of New Brunswick. [Online]. Available: <https://www.unb.ca/cic/datasets/ids-2018.html> SMARTPANTS PRESERVED 328 SMARTPANTS PRESERVED 329

- [L. Engelen, V. Rimmer, and W. Joosen, "Troubleshooting an Intrusion Detection Dataset: the CICIDS2017 Case Study," in *Proc. IEEE Security and Privacy Workshops*, 2021, pp. 7–12.](#)
- [N. V. Chawla et al., "SMOTE: Synthetic Minority Over-sampling Technique," *JAIR*, vol. 16, pp. 321–357, 2002.](#)

[Verify page numbers and the Engelen venue before submission.](#)